

Marian-Andrei RIZOIU

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CURRICULUM VITAE

About me

Marian-Andrei Rizoiu is a Professor leading the [Behavioral Data Science lab](#) at the University of Technology Sydney. His interdisciplinary research crosses computer and social sciences, blending psycholinguistics, digital communication and stochastic modelling to understand human attention dynamics in the online environment, the emergence of influence and opinion polarization. Marian-Andrei is the recipient of the prestigious Excellence Award and Academic of the Year at the [2023 Australian Defence Industry Awards](#) and a finalist of the distinguished [Australian Museum Eureka Prizes](#) for Outstanding Science in Safeguarding Australia. He currently leads grants worth \$5.9 million from the Commonwealth of Australia to detect and model the spread of mis- and disinformation and its weaponized counterparts – information and influence operations.

Marian-Andrei's research receives funding from selective funders such as Meta (Facebook) Research, Defence Science and Technology Group (DSTG), The Department of Home Affairs and the Defence Innovation Network. In addition, he publishes in the most selective venues, such as the PNAS, PLOS ONE, PLOS Computations Biology, WWW, NeurIPS, IJCAI, and CIKM. As a result, his work has received significant media attention—including [Bloomberg Business Week](#), [Nature Index](#), [BBC](#), and [World Economic Forum](#).

Marian-Andrei disseminates his research to the broader public by regularly contributing to [The Conversation](#). He also leverages his research to real societal impact; he was an expert for the NSW government's Defamation Law Reform or provided evidence for the Australian Federal Senate inquiry into media diversity.

Research interests

- Stochastic behavioural modelling; online information diffusion; online popularity modelling.
- Detecting disinformation campaigns and malicious activity, information warfare, online influence and disruptive socialbots in democratic processes.
- Labour markets in automation age, measuring skill similarity, labour upskilling, job ads data.

Employment history

- 01/2024 – current **Associate Professor**, Data Science Institute, University of Technology Sydney
Research: information diffusion models, mis-/disinformation detection, online influence operations
Supervision: 2 postdoc, 5 PhD students, 2 Masters/Honours students.
- 2021 – 2023 **Senior Lecturer**, Data Science Institute, University of Technology Sydney
Research: information diffusion models, mis-/disinformation detection, online influence operations
Supervision: 1 postdoc, 7 PhD students, 2 Masters/Honours students.
- 2019: **Visiting Professor**, Jean Monnet University & CNRS Hubert Curien lab, Saint-Etienne, France
Research: dynamics of online polarisation
- 2019 – 2021: **Lecturer**, Faculty of Engineering and IT, University of Technology Sydney
Research: information diffusion models, online popularity prediction models, stochastic processes
Supervision: 1 postdoc, 4 PhD students, 3 Masters/Honours students/Research Assistants.
- 2016 – 2019 **Research Fellow**, College of Engineering and Computer Science, Australian National University
Research: information diffusion models, online popularity prediction, stochastic point processes.
Teaching: Document Analysis (convener), Advanced Databases and Data Mining.
Supervision: 4 PhD students (2 as main supervisor), 3 Masters/Honours students.
- 2014 – 2016 **Research Scientist**, Optimisation Research Group, National ICT Australia (NICTA)
Research: online privacy, big social data analytics, data mining.
Supervision: 2 PhD students, 1 Masters/Honours student.
- 2013 – 2014 **Postdoctoral Fellow**, ERIC laboratory, Lumière University Lyon, France.
Research: ontology construction from text, knowledge graphs evolution for brand management.
Teaching: Machine Learning, Software Development Methodologies, Data Mining, OOP

Supervision: 4 Honours theses, 9 industrial internships.

Education background

- 2009 – 2013 **PhD in Computer Science**, Lumière University Lyon, France.
(June 24th 2013) “Semi-supervised structuring of complex data”, supervision S. Lallich and J. Velcin.
- 2008 – 2009 **Masters Degree in Data Mining and Knowledge Engineering**,
(double diploma) Polytechnic School, Nantes University, France.
- 2004 – 2009 **Engineer Degree in Systems and Computer Engineering**
School of Computer Science, Polytechnic University of Bucharest, Romania.

Grants, Projects and Consulting – AU\$11.8M total funding (\$4.69M as lead-CI); Sample:

- 2023-2025: Defence Innovation Network. “**Tracking Policy Influencing Pathways in Asia-Pacific**”, AU\$500K, Lead-CI. Understanding the narratives and vectors of policy influence.
- 2023: Defence Science and Technology, IWSS. “**NARRATE - Narrative and Relationship Real-time Awareness Topical Explorer**”, AU\$100k, Lead-CI. Identify the narratives of state-backed agents.
- 2022-2025: Department of Home Affairs. “**Disinformation Defence Initiative: Delivering tools and analysis to fight the growing threat of disinformation for Australia**”, AU\$1M, Lead-CI. Linking online misinformation to domestic extremism, and the importation of foreign misinformation narratives.
- 2022-2025: [Industrial Doctorate Program](#). “Supervision of PhD students embedded into Australian companies”, AU\$30K, Lead-CI. Student supervision, commercial-oriented research.
- 2021 – 2023: Defence Science and Technology (DST). “**Forensic analysis and real-time detection of dis-information campaigns**”, AU\$300K, Lead-CI. Real-time detection of disinformation campaigns using multi-faceted social media analysis.
- 2021: [Polish National Science Centre OPUS grants](#), “**Streaming social data**”, AU\$313K, CI. Model the dynamics of social interactions using data streams and complex networks theory.
- 2021: [UTS Cross-Faculty Scheme](#), “**The dynamics of disinformation across traditional and social media**”, AU\$37K, Lead-CI. Understand the dynamic interaction between the traditional and social media ecosystems that results in the flow of disinformation and problematic content.
- 2020-2021: [Facebook Research grants](#), “**Using computational modelling of user behaviour and machine learning to counter the diffusion of hate speech across social media**”, US\$60k, CI. Use ethnographic methods to identify and monitor a number of persona associated with hate speech diffusion targeting vulnerable populations.
- 2019 – 2020: National Security College’s Green policy grants, “**Tracking Disinformation Campaigns Across Terrains: Implications for Policy**”, AU\$50K, CI. Quantify the scale of the problem of disinformation in order to co-design responses with policy partners.
- 2019 – 2020: UTS Cross-Faculty Scheme, “**SocialSense: Making sense of the opinions and interactions of online users**”, AU\$20K, CI. Study the diffusion and polarization of opinions online, mixing an ethnographic approach with computational modelling of behaviour.
- 2019: Science and Industry Endowment Fund, “**Adaptive skills taxonomy to enable labour market agility**”, AU\$350K, CI. Understand the change of demand and supply of skills in a timely and efficient manner in order to provide information to inform employment decisions.
- 2019 Industrial consulting with a merger/acquisition. AU\$10.5k, Lead-CI.
- 2018 ANU Social Science Cross-College Grants, “**Advanced tools and methods for analysing the role and influence of bots in social media**”, AU\$50K, Lead-CI. The impact of automation in the form of socialbots on deliberative democracy – how socialbots hijack the public discourse.
- 2018 ANU Social Science Cross-College Grants, “**Identify Hate Speech and Predict Mass Atrocities**”, AU\$30K, Lead-CI. Can “hate speech” be reliably measured to predict political violence?

Invited talks

I gave more than 40 invited talks to varied industrial, academic and government stakeholders, such as to Silicon Valley tech companies (Facebook, Netflix), international universities and research labs, and Australian governmental institutions. See a selection:

- 09/2021 Occupation transition recommender systems. **LinkedIn Tech Talk Series**.
- 11/2020 Career transitions and managing labour supply and demand. **Reserve Bank of Australia**.

- 10/2020 Fighting the proliferation of problematic speech on online social networks. **Facebook Content Policy and Moderation**, Palo Alto, California, USA.
- 09/2020 Skill shortages and worker transitions. **Victoria government, Department of Jobs, Precincts and Regions**, Melbourne, Australia.
- 04/2020 Career Development Recommender System & COVID implications for employment. **New South Wales Treasury**, Sydney, Australia.
- 04/2020 Skills and employment in times of COVID-19. **Department of Education, Skills and Employment, Australian Federal Government**, Canberra, Australia.
- 11/2019 Information warfare: socialbots and Russian online trolls in the US elections. Invited talk at the **Defence Science and Technology (DST)**, Adelaide, Australia.
- 05/2019 #DebateNight: Role of Twitter Socialbots During US Presidential Debate. Invited talk at the **Computer Society of the IEEE ACT Section**, Canberra, Australia.
- 07/2019 Studying Online Video Popularity with Computational Models. **Sage Research Methods**.
- 06/2018 Hawkes Intensity Processes for modelling online popularity and virality. Invited talk at **Facebook Core Research**, Palo Alto, California, USA.
- 06/2018 User engagement with online video and the unpredictability of online popularity. Invited talk at **Netflix Research**, San Jose, California, USA.
- 05/2018 Scalable influence estimation from online information diffusions. Research visit at the **Max Plank Institute for Software Systems**, Kaiserslautern, Germany.
- 03/2016 Evolution of Privacy Loss in Wikipedia. Invited talk, March session of the Monthly **Wikimedia Research Showcase**, San Francisco, USA.

Media engagement and outreach

The Conversation – I wrote 7 articles for The Conversations, with more 161,000 reads (source: The Conversation, 04/2023), reposted on websites like [World Economic Forum](#) and [BBC](#) and with significant social media attention.

- 02/2023 [Can ideology-detecting algorithms catch online extremism before it takes hold?](#)
- 09/2022 [A huge LinkedIn study just showed which connections are better when searching for a job](#)
- 08/2021 [How AI can help choose your next career and stay ahead of automation](#)
- 04/2021 [We spent six years scouring billions of links, and found the web is both expanding and shrinking](#)
- 03/2020 [Coronavirus infecting Australian jobs: vacancy rates down since early February](#)
- 12/2019 [Robot career advisor: AI may soon be able to analyse your tweets to match you to a job](#)
- 10/2019 [Can hiding likes make Facebook fairer and rein in fake news? The science says maybe](#)

Radio

- 09/2021 **2ser radio** – [Streamers go dark over hate raids](#)
- 09/2021 **2ser radio** – [How AI Is Helping Aussies Avoid Automation](#)
- 05/2021 **Radio Adelaide** Breakfast show – [The Internet Is Massive and Small](#)
- 10/2020 **Syn Media** – [News Talks #11: How social media spreads misinformation](#)
- 11/2019 **2ser radio** – [Facebook and Fake News](#)
- 10/2019 **ABC radio** – Hiding likes make Facebook fairer
- 10/2019 **Radio Adelaide** – [Hiding the Number Of Likes On Social Media](#)

Media coverage – My research received significant media attention – 16 of my papers were covered by 84 news and blogs, 1 policy document, 4 Wikipedia pages and 878 mentions on social media (source: Altmetric 25/05/2021). Selected examples (see [full list of media coverage here](#)):

- 08/2021 **World Economic Forum** – [Fancy a career change? AI could help you decide your next move](#)
- 08/2021 **LinkedIn News Australia** – [Want to pivot careers? Ask AI](#)
- 05/2021 **World Economic Forum** – [What these researchers discovered after studying the internet for 6 years](#)

05/2021 **RT News** – [60-70% of world's attention focused on just 10 online domains, according to latest research](#)

05/2021 **Science Alert** – [The Same Handful of Websites Are Dominating The Web And That Could Be a Problem](#)

05/2021 **Canaltech Brasil** – [Internet mundial cresce ao mesmo tempo em que está cada vez menor; entenda](#)

03/2021 **ABC News** – [Facebook promised to ban anti-vaxxers. But are still up and they've been selling t-shirts](#)

02/2020 **Bloomberg Businessweek** – [The Best Way to Change Your Job Is to Focus on Your Personality](#)

02/2020 **Nature Index** – [Scientists are curious and passionate and ready to argue](#)

02/2020 **BBC** – [How your Twitter feed could help find your dream job](#)

Community Contributions

03/2021 My paper "Layoffs, Inequity and COVID-19" [8] was used as evidence in the [Senate inquiry into media diversity](#) by Marcus Strom, the president of Media Entertainment and Arts Alliance ([submission 26](#)).

10/2020 I consulted the Facebook's Content Policy and Regulation team, who is looking for means to automatise their content moderation.

01/2020 I served as an expert for the **Council of Attorneys General of New South Wales (NSW)** for the [defamation law reform](#), helping the NSW government develop their understanding of online social networks and internet service providers. The law reform proposal voted into law in August 2020.

Honours, prizes and awards

09/2024 (finalist) [Australian Museum Eureka Prizes](#) for Outstanding Science in Safeguarding Australia

09/2023 **Excellence Award and Academic of the Year Award**, [2023 Australian Defence Industry Awards](#).

11/2020 **PitchFest award for disinformation prototype**. NSW Smart Sensing Network & Klarrio Ltd.

03/2019 [CNRS IDEXLYON](#) award for research excellence. French National Research Agency.

04/2018 **Travel Awards** – to travel to ECML-PKDD'15, WWW'18. ANU Early Career Researchers grants

01/2015 **NVIDIA Titan GPU**. NVIDIA GPU Awards Program.

11/2012 **Best student paper award**. International Conference on Tools with Artificial Intelligence, 2012.

Professional service

Area chair / PC member Area Chair: ECML-PKDD'21 '22 '23;
PC: WWW'20'21'22, AAAI'20'22, AAAI'19'21, ICWSM'19-'22, KDD'22

Reviewing Journal of Machine Learning Research, Computational Intelligence, Transactions on Information Systems, Journal Transactions on Knowledge Discovery from Data

Conference organisation 10/2012: Discovery Science 2012 and Algorithm Learning Technology 2012, Lyon.
09/2011: Web Intelligence 2011 and Intelligent Agent Technology 2011, Lyon.

Selected recent publications

Publications summary: 56 peer reviewed publications, 1825 citations, H-index 25 (source: *Google Scholar* 01/2024). Full publication list at <https://www.behavioral-ds.science/publication/>

- [1] Schneider, P. J., & Rizoïu, M.-A. (2023). *The effectiveness of moderating harmful online content*. **Proceedings of the National Academy of Sciences (PNAS)**, 120(34), 1–3. <https://doi.org/10.1073/pnas.2307360120>
- [2] Calderon, P., Ram, R., & Rizoïu, M.-A. (2024). Opinion Market Model: Stemming Far-Right Opinion Spread using Positive Interventions. In *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM'24)* (pp. 1–12). Retrieved from <http://arxiv.org/abs/2208.06620>
- [3] Kong, Q., Calderon, P., Ram, R., Boichak, O., & Rizoïu, M.-A. (2023). Interval-censored Transformer Hawkes: Detecting Information Operations using the Reaction of Social Systems. In *Proceedings of the ACM Web Conference 2023 (WWW'23)*, pp. 1813–1821. New York, NY, USA: ACM. <https://doi.org/10.1145/3543507.3583481>
- [4] Rizoïu, M.-A., Soen, A., Li, S., Calderon, P., Dong, L., Menon, A. K., & Xie, L. (2022). *Interval-censored Hawkes processes*. **Journal of Machine Learning Research** <http://arxiv.org/abs/2104.07932>
- [5] Kong, Q., Booth, E., Bailo, F., Johns, A., & Rizoïu, M.-A. (2022). Slipping to the Extreme: A Mixed Method to Explain How Extreme Opinions Infiltrate Online Discussions. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM'22)*, 16(1), 524–535. <https://doi.org/10.1609/icwsml.v16i1.19312>
- [6] Dawson, N., Williams, M.-A., & Rizoïu, M.-A. (2021). *Skill-driven recommendations for job transition pathways*. **PLOS ONE**, 16(8), e0254722. <https://doi.org/10.1371/journal.pone.0254722>
- [7] McCarthy, P. X., Gong, X., Eghbal, S., Falster, D. S., & Rizoïu, M.-A. (2021). *Evolution of diversity and dominance of companies in online activity*. **PLOS ONE**, 16(4), e0249993. <https://doi.org/10.1371/journal.pone.0249993>
- [8] Dawson, N., Molitorisz, S., Rizoïu, M.-A., & Fray, P. (2021). *Layoffs, inequity and COVID-19: A longitudinal study of the journalism jobs crisis in Australia from 2012 to 2020*. **Journalism**. p. 146488492199628

- [9] Kong, Q., **Rizoiu, M.-A.**, & Xie, L. (2020). *Describing and Predicting Online Items with Reshare Cascades via Dual Mixture Self-exciting Processes*. In Proceedings of the 29th ACM International Conference on Information & Knowledge Management (**CIKM'20**), pp. 645–654, New York.
- [10] Kong, Q., **Rizoiu, M.-A.**, & Xie, L. (2020). *Modeling Information Cascades with Self-exciting Processes via Generalized Epidemic Models*. In ACM International Conference on Web Search and Data Mining (**WSDM'20**). Houston, Texas.
- [11] Zhang, R., Walder, C., & **Rizoiu, M.-A.** (2020). *Variational Inference for Sparse Gaussian Process Modulated Hawkes Process*. In the AAAI Conference on Artificial Intelligence (**AAAI'20**), 34(04), 6803–6810.
- [12] Zhang, R., Walder, C., Bonilla, E. V., **Rizoiu, M.-A.**, & Xie, L. (2020). *Quantile Propagation for Wasserstein-Approximate Gaussian Processes*. In *Conference on Neural Information Processing Systems (NeurIPS'20)*.
- [13] Kern, M. L., McCarthy, P. X., Chakrabarty, D., & **Rizoiu, M.-A.** (2019). *Social media-predicted personality traits and values can help match people to their ideal jobs*. Proceedings of the National Academy of Sciences (**PNAS**), 201917942.
- [14] Zhang, R., Walder, C., **Rizoiu, M.-A.**, & Xie, L. *Efficient Non-parametric Bayesian Hawkes Processes*. In: International Joint Conference on Artificial Intelligence (**IJCAI'19**), Macao, China, 2019.
- [15] **Rizoiu, M.-A.**, Mishra, S., Kong, Q., Carman, M., & Xie, L. *SIR-Hawkes: Linking Epidemic Models and Hawkes Processes to Model Diffusions in Finite Populations*. In: Proceedings of International Conference on World Wide Web (**WWW '18**), Lyon, France, pp. 1–9, 2018.
- [16] **Rizoiu, M.-A.**, Graham, T., Zhang, R., Zhang, Y., Ackland, R. J., & Xie, L., *#DebateNight: The Role and Influence of Socialbots on Twitter During the 1st U.S. Presidential Debate*. In International AAAI Conference on Web and Social Media (**ICWSM'18**), pp. 1–10, 2018.
- [17] **Rizoiu, M.-A.**, Xie, L., Sanner, S., Cebrian, M., Yu, H., & Van Hentenryck, P., *Expecting to be HIP: Hawkes Intensity Processes for Social Media Popularity*. In: Proceedings of International Conference on World Wide Web (**WWW '17**), Perth, Australia, pp. 735-744, 2017.
- [18] **Rizoiu, M.-A.**, Xie, L., Caetano, T., & Cebrian, M., *Evolution of Privacy Loss in Wikipedia*. In: Proc. International Conference on Web Search and Data Mining (**WSDM '16**), San Francisco, USA, pp. 215-224, February 2016.